Texas A&M University
Department of Nuclear Engineering
NUEN 406 Nuclear Engineering Systems and Design
Fall 2025

Statement of Design Philosophy and Integration

Philosophy

Objectives:

As a major component of their Baccalaureate education, every undergraduate Nuclear Engineering student will have a firm grounding in the:

a. principles, practices and analytical tools of each stage of engineering design to include problem definition, conceptualization, design evaluation and communication of the design.
b. fundamentals of design and economic decision making, cost engineering, reliability engineering, and design for value and quality.
c. comprehensive evaluation of design option selection based on performance, economic, reliability and esthetic factors.
d. design of products and processes in the broader contexts of engineering ethics, societal needs and preferences, product life cycle, systems engineering, human factors, and environmental protection.
e. the role of standards, specifications, quality engineering, risk assessment and the protection of intellectual properties in engineering design.

Wherever possible, open-ended problems will be addressed in conjunction with coursework, which require innovative conceptualization and hands-on experience and which must be communicated in written and oral reports. These problems will demand that students use their knowledge in an integrated manner and that they use good engineering judgment and insight to produce original results of substantial value. These problems will be designed to help the student gain an appreciation for the fact that most engineering design problems do not have unique solutions or solution paths. Nor can they always be fully verbalized or analyzed. Furthermore, they must not only meet functional design requirements but also accommodate a variety of constraints, such as those imposed by environmental, sociological, political, ethical, and economic, health and safety, manufacturability, constructability, global and cultural considerations.

Integration into the Undergraduate Curriculum

These principles of design are integrated in the capstone courses NUEN 406 and NUEN 410, which were developed to give Nuclear Engineering seniors background in design methodologies and the interplay of design objective functions to optimize performance, cost, reliability and total quality. The courses also integrate experience-based training in team dynamics, communications, and project planning and execution.

The team projects are realistic contemporary problems at the interface between research and engineering development. These problems require both information synthesis and analysis in order to address both the product and process design aspects of the problem. In some problems, in-depth diagnostics, root-cause analysis, and failure modes and effects analysis are required under the mentorship of one or two faculty members. Hence, the students gain practical experience in team dynamics, leadership skills, and communication.

Lectures are designed to provide a fundamental understanding of concept engineering, decision theory, engineering economics, engineering statistics, risk analysis, total quality, and systematic design option selection against specific design objectives and requirements. Homework and project assignments are designed to integrate these principles in solving open-ended, realistic engineering problems.

Project assignments require that student teams do sufficient background research and information analysis in their assigned problems to develop a comprehensive project plan with implementation schedules and cost estimates. ***Each project must contain a reactor physics/radiation component and three to four other nuclear-related components of the team's choosing so that each team member takes the lead in one technical component. In addition, there must be an economic analysis***.

In NUEN 406, each project team is required to prepare and present their Project Plan. At the end of the spring semester in NUEN 410, each team will submit a formal project report and present their design approach, findings and conclusions to the department faculty, students and external evaluators.

DESIGN PROJECT POLICIES AND PROCEDURES

Teams: Teams will be formed to exercise team building skills, conduct senior projects under NUEN 410, work classroom exercises (such as formal design reviews), and prepare and present the project report.

The team will select a team captain. The designated team captain will be responsible for scheduling meetings with the technical advisor and course instructor as appropriate, notifying the technical advisor and course instructor when issues arise.

Technical Advisor: Each team will select a technical advisor(s) for their project. The technical advisor should have expertise in one or more areas of the project and be willing to advise the team through the NUEN 406 and 410 semesters. The advisor may be a faculty member in or outside of the Nuclear Engineering department, an employment supervisor or other qualified personnel. If the advisor is outside of the university, the team must also secure a faculty advisor within the department to ensure that all academic requirements are met.

The team is responsible for motivating the team-advisor relationship. The team must actively seek input from the advisor, request regular meetings and provide updates to the advisor.

Since the project deliverables will require the signature of the technical advisor, advance time must be given for the advisor to review all necessary materials. Do not expect the advisor to be available immediately prior to deadlines!

Project Plan Review: The NUEN Advisory Council will review the Project Plans and provide comment for consideration in NUEN 410. This review is independent from the course instructor's grading of the projects. Grading will be performed by the instructor, without direct input from the Advisory Council.

Grading: Grading of the Project Plans will be as listed below:

Organization and Appearance	10%
Grammar and Exposition	10%
Content and Technical Approach	65%
40% for Scoping Analysis	
40% for Approach	
20% for remainder of content	
Review/Approval by Advisor	10%
Submittal by Due Date	5%

*1 bonus point will be added to the Project Plan grade for each day early that the plan is submitted.

Project Plan Content: Each team will prepare the Project Plan in NUEN 406 according to the following outline. The length should be about 10 pages of text in length, single-spaced, 12-point, exclusive of the title page, approval page, figures and tables, references and appendices.

Title Page: Include team identification, names and dates.

Approval Page: Same as the title page with signature lines for all technical advisors.

Executive Summary: Give in one or two paragraphs on a separate page, an abbreviated statement of the background, the objectives of the project, the approach intended, the resources and the time required.

Introduction: Include answers to as many of the following questions as appropriate:

- What does the reader need to know to understand the project?
- What is the background leading to the project?
- Why is the project being done and what is being attempted?
- How is it done today?
- If the approach is different, why is it different?
- If the project is successful, what will be the significance?

Objectives: Describe in 2 to 4 lines the principal objectives.

Scoping Analysis: Document the scoping analysis performed in each technical area that establishes the feasibility of the design concept. Discuss:

- Assumptions
- Constraints and Engineering Standards to be accommodated
- Hand calculation results (order of magnitude analysis)
- Initial survey results (materials selection, regulatory requirements, etc.)
- Selection of codes to be used/status of codes that were written for use in NUEN 410
- Design concept viability

Approach: Describe how the team intends to carry out the project in NUEN 410 account for the following factors, as appropriate:

- work elements (major subtasks)
- project schedule (milestones, completion dates for each, Gantt chart)
- project organization (who is assigned what?)
- facilities and equipment required (if appropriate)
- cost estimates (as appropriate)
- other (specify)
- discussion (discuss how the approach and methods of analysis will successfully achieve the objectives)

References

Appendices